Deepan Chandrasekaran

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PROFESSIONAL SUMMARY

Computer Science undergraduate (CGPA 9.15, graduating April 2027) with proven ability to design, develop, and deploy secure, high-quality software solutions across backend systems, microservices, and full-stack applications. Experienced in Java, Python, and JavaScript with hands-on exposure to CI/CD practices, cloud infrastructure, agile development, and data pipeline engineering. Passionate about building resilient, accessible technology that creates real-world impact for diverse communities.

EDUCATION

B.Tech in Computer Science and Engineering at KL University *Expected: April 2027*

CGPA: 9.15 / 10 | Vijayawada | Coursework: Data Structures & Algorithms, Operating Systems, DBMS, OOP, Computer Networks, Software Engineering

WORK EXPERIENCE

Software Developer (Freelance) – Local Apparel Manufacturer

2023 | Freelance

- Designed and delivered a full-stack web application against defined client requirements — responsive React frontend with a backend order-management system — applying the full software development life cycle from analysis through deployment, increasing sales by 8–10%
- Engineered a real-time 3D product-preview feature using Three.js; collaborated directly with the client to interpret nonfunctional requirements (performance, accuracy) and align the solution to production specifications
- Resolved third-party API integration defects under delivery constraints; gathered and acted on end-user feedback to continuously improve software reliability and usability

CERTIFICATIONS

Oracle Cloud Infrastructure 2025 – Generative AI Professional at Oracle University

Oct 2025 – Oct 2027

Oracle AI Vector Search Professional at Oracle University

Oct 2025 – Oct 2027

SKILLS

Languages: Java, Python, JavaScript, C/C++

Frameworks & Libraries: Spring Boot, FastAPI, React, PyTorch, Three.js, TailwindCSS, Sentence Transformers

Tools & DevOps: Git, GitHub, Docker, Linux, AWS (basics), CI/CD fundamentals, Vercel, Ollama

Data & Storage: SQL, PostgreSQL, ChromaDB, FAISS, Big Data concepts, Data Warehousing fundamentals

Concepts: REST APIs, Microservices, Distributed Systems, Application Resiliency, OOP, Agile SDLC, Secure Coding, Computer Vision, Semantic Search

PROJECTS

LaRa – Resilient Distributed Microservices Platform *Jan 2026 – Present*

Tech: *Python, FastAPI, WebRTC VAD, Faster-Whisper, Docker, Microservices*

- Social impact platform built for **neurodiverse and specially-abled children** — awarded **3rd Place at RAMPAGE V26 National Hackathon, KL-H**
- Architected fault-tolerant containerised microservices using FastAPI with async endpoints (/health, /start, /stop); applied service isolation and resiliency patterns to ensure stable, production-grade operation
- Built a secure, offline-capable real-time speech pipeline (Faster-Whisper + WebRTC VAD) with no cloud dependency — ensuring data privacy and application resiliency for sensitive environments; implemented modular session-state management and analytics modules across the SDLC

ResumeAnalyse – Semantic Data Pipeline & Full-Stack Web Application *Dec 2025 – Feb 2026*

Tech: *Sentence Transformers, ChromaDB, Ollama, Mistral LLM, React, Python, REST APIs*

- Designed and implemented a high-accuracy semantic vector search data pipeline using embedding models and ChromaDB, achieving 90%+ matching relevance — demonstrating proficiency in data integrity and retrieval accuracy at scale
- Developed secure RESTful backend services consumed by a React frontend; automated insight generation reduced manual analysis time by 50%, directly applying automation to reduce toil in a data-driven workflow

ISL Sign Language Recognition – End-to-End ML Engineering Pipeline *Aug 2025 – Nov 2025*

Tech: *PyTorch, CNNs, Python, NVIDIA A100 / H100 / T4*

- Engineered a full SDLC pipeline — from large-scale data ingestion (42,000 samples, architected to 1M+) to training and real-time inference — achieving 99.8% validation and 90% test accuracy on NVIDIA A100, H100, T4 GPUs
- Optimised system throughput to 45 FPS on edge devices; applied performance engineering and scalable architecture principles consistent with big data and distributed system design best practices

CODING PROFILES

LeetCode: https://leetcode.com/u/kl2300032731/ **CodeChef:** https://www.codechef.com/users/klu2300032731